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SECTION 2

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TUTORIAL ARTICLE

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GROUP 2

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1.0 Main Problem

In this research, PT. MPM is a packaging manufacturing company. PT. MPM does not have a proper system or application to monitor the performance of its core business processes, particularly in sales and production operations. Besides, its management may still rely on manual reporting, scattered and disorganized data sources. As a result, top management met trouble on viewing the key performance indicators (KPIs) of their company such as sales achievement, production efficiency, production capacity and Overall Equipment Effectiveness (OEE). The lack of efficient and real-time performance information led to poor decision-making processes and reduced the company's ability to respond quickly to operational and market changes. In short, PT MPM met challenges in improving operational efficiency and maintaining competitiveness within the manufacturing industry.

2.0 Main Solution

A Business Intelligence (BI) solution was suggested to address these challenges using two interactive dashboards which are the Sales Performance Dashboard and the Production Performance Dashboard. Specifically, the Sales Performance Dashboard enables top management to overview the sales revenue, sales achievement, and customer analysis for future forecasting decisions. Conversely, the Production Performance Dashboard evaluates manufacturing metrics, machines performance, downtime analysis and their Overall Equipment Effectiveness (OEE). The system integrates data from multiple departments and cleans them through an ETL (Extract, Transform, Load) process. Then, the processed data will be stored in a centralized data warehouse.

The two dashboards were developed using Google Data Studio following the Waterfall Software Development Life Cycle (SDLC) methodology. The dashboards would provide the top management with clear, accurate and real-time information about the company performance using charts, scorecards, tables and reports. This enables data-driven decision-making, improves performance monitoring and enhances overall operational efficiency. The developed BI dashboard was evaluated through alpha and beta testing. It successfully achieved an overall user satisfaction score of 82% based on the article.

3.0 User Stories

3.1 User Story 1 – Sales Staff

As a Sales Staff, I want to upload and update our sales records such as product categories, quantities sold and revenue forecast so that the company can accurately track the overall sales performance against actual daily transactions.

3.2 User Story 2 – Manufacturing Staff

- As a Manufacturing Staff, I want to update our daily production records like our expected versus actual output of products, so that management has a clear picture of the daily production line's productivity.
- As a Manufacturing Staff, I want to record machine performance details, like machine availability and product quality, so that the system can provide accurate Overall Equipment Effectiveness (OEE) metrics for every machine.

3.3 User Story 3 – Inventory Staff

As an Inventory Staff, I want to always maintain and update our material stock levels in the system so that the production team could get the real-time information they need, to adjust their manufacturing schedules without delays.

3.4 User Story 4 – Technical Staff

As a Technical Staff, I want to input new product details and packaging specifications into the system so that everyone across the manufacturing and sales teams is using the same standardized reference materials.

3.5 User Story 5 – Data Analyst

- As a Data Analyst, I want to extract, transform and load (ETL) the raw data provided by various departments so that all the data is clean, properly formatted and ready for analysis.
- As a Data Analyst, I want to manage and maintain the data warehouse so that our historical and current operational data remains secure and able to provide quality data for reporting tools without downtime.
- As a Data Analyst, I want to build and generate interactive performance dashboards so that the top management team can visualize complex data metrics without needing to manually check and analyse through raw spreadsheets.

3.6 User Story 6 – Top Management

- As Top Management, I want to interact with the performance dashboards so that I can quickly monitor metrics like sales revenue and company KPIs to make strategic business decisions.
- As Top Management, I want to export or print specific views from the dashboard so that I can easily share important business insights and performance summaries with other stakeholders.

4.0 ERD

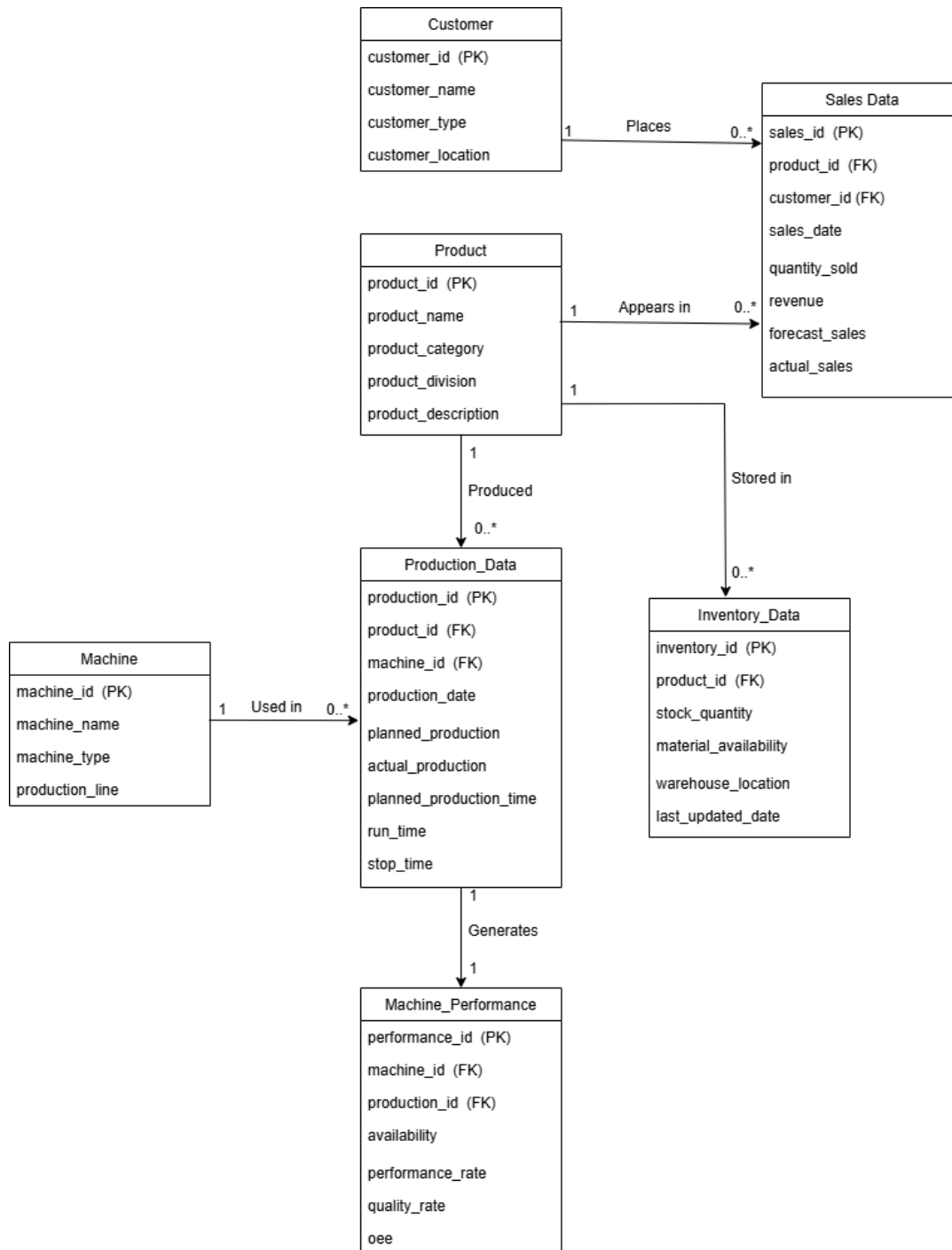


Figure 1 Entity Relationship Diagram (ERD) of PT. MPM BI Dashboard System

Figure 1 illustrates the Entity Relationship Diagram (ERD) for the PT. MPM Business Intelligence Dashboard System. The ERD represents the logical database structure required to support the collection, storage, and analysis of sales, production, machine performance, and inventory data. The database

consists of seven main entities: Customer, Product, Sales Data, Production Data, Machine, Machine Performance, and Inventory Data.

The **Customer** entity stores customer-related information such as customer ID, customer name, customer type, and customer location. A customer can generate multiple sales transactions; therefore, a one-to-many relationship exists between Customer and Sales Data.

The **Product** entity contains information about the products manufactured and sold by the company, including product category, division, and description. Since a product can appear in multiple sales transactions, production records, and inventory records, the Product entity is linked to Sales Data, Production Data, and Inventory Data through one-to-many relationships.

The **Sales Data** entity records sales transactions such as quantity sold, revenue, forecast sales, and actual sales. This information is used to evaluate sales performance and compare forecasted results with actual business outcomes.

The **Production Data** entity stores manufacturing-related information including planned production, actual production, planned production time, run time, and stop time. Each production record is associated with a specific product and machine.

The **Machine** entity contains information about production machines used in the manufacturing process. One machine can be involved in multiple production activities, resulting in a one-to-many relationship between Machine and Production Data.

The **Machine Performance** entity stores machine performance indicators including availability, performance rate, quality rate, and Overall Equipment Effectiveness (OEE). Each production record generates one corresponding machine performance record; therefore, a one-to-one relationship exists between Production Data and Machine Performance.

The **Inventory Data** entity maintains information regarding stock quantity, material availability, warehouse location, and inventory update dates. This entity supports inventory monitoring and production planning activities.

Overall, the ERD provides the foundation for storing operational data that will later be processed through the ETL process and visualized in the Business Intelligence dashboards for management decision-making.

5.0 Use Case Diagram

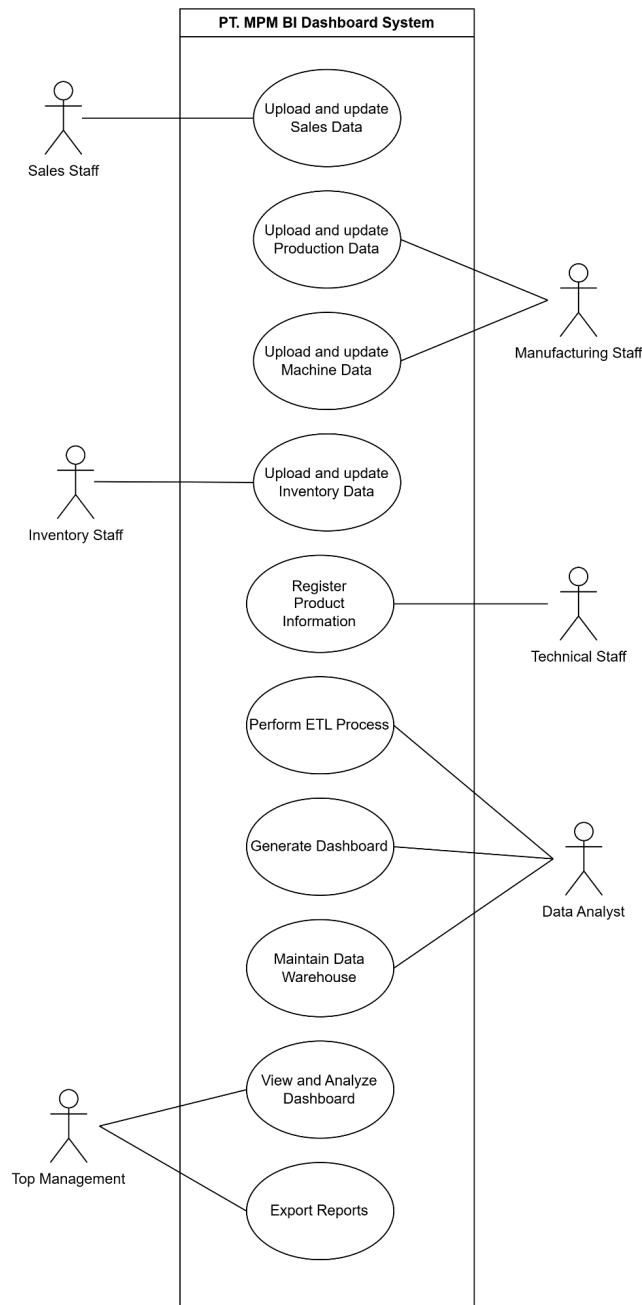


Figure 2 Use Case Diagram of PT. MPM BI Dashboard System

Figure 2 shows the use case diagram for the PT. MPM Business Intelligence (BI) Dashboard System. This use case diagram outlines the interactions between the actors and the system's core functional modules which are the use cases. There are six distinct actors representing different departmental roles within the company. They are Sales Staff, Inventory Staff, Manufacturing Staff, Technical Staff, Top Management and the Data Analyst.

Based on the use case diagram, there are a few departmental staff who are responsible for supplying the necessary data needed by the system. Firstly, the Sales Staff has to upload and update the sales data, while the inventory data is managed by the Inventory Staff. Next, the production records and machine performance data are uploaded and updated by the Manufacturing Staff. All the product information is registered by the Technical Staff. Once these data are generated, they will be processed by the Data Analyst. The data analyst's responsibility is to perform the Extract, Transform, Load(ETL) process structuring the data, maintain the data warehouse and generate the BI dashboards. Finally is the Top Management who will be using the visual generated to make strategic decisions and forecasting.

6.0 System Architecture

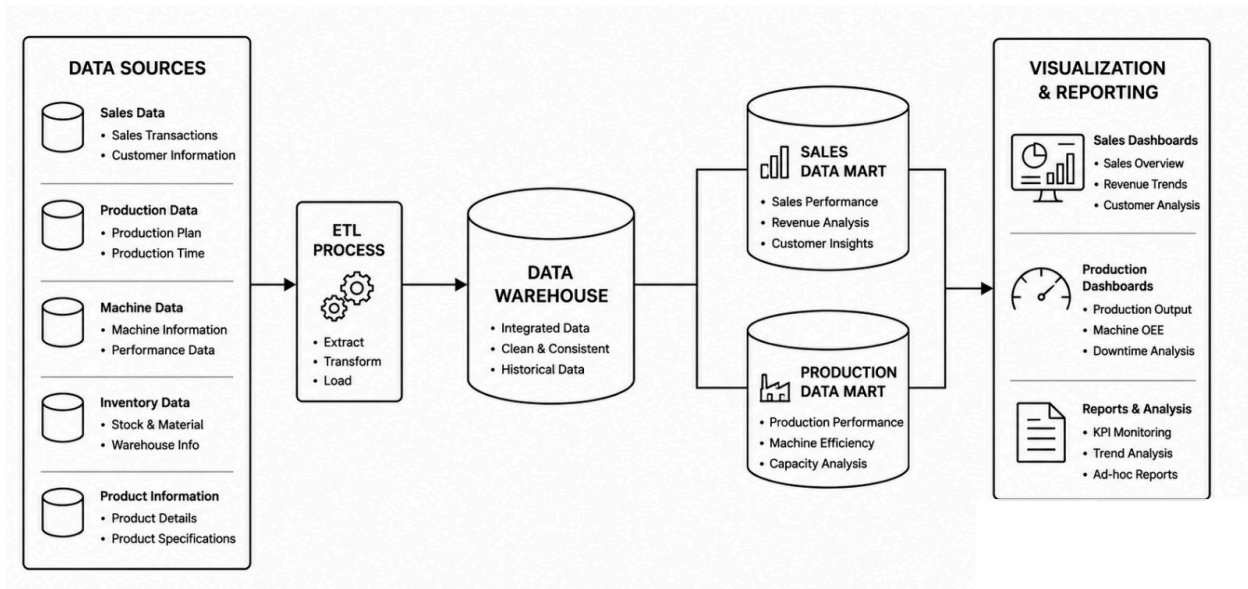


Figure 3 System Architecture of the PT. MPM BI Dashboard System

Figure 3 shows the system architecture for the PT. MPM Business Intelligence Dashboard System. It illustrates the systematic data flow from raw data collection to the final visualization for top management. First of all, separated raw data includes sales transactions, production data, machine performance metrics, inventory stock and product information are gathered within the Data Sources layer. Then, these inputs are ingested into the ETL Process phase where data is extracted from the respective departments, transformed to ensure it is clean and formatted and then loaded into the central Data Warehouse. The data warehouse which utilizes Google Sheets as its primary tool, stores this integrated historical data using a galaxy schema. From the central warehouse, the architecture segments the data into dedicated Data Marts, specifically a Sales Data Mart and a Production Data Mart, to organize the metrics for targeted departmental analysis. Finally, this highly structured data is fed into the Visualization & Reporting layer. By using Google Data Studio as the primary BI platform, the system generates comprehensive Sales Dashboards, Production Dashboards and additional analytical reports to facilitate KPI monitoring, trend analysis and data-driven decision-making.

7.0 Galaxy Schema Design

To support the analytical and reporting needs of PT. MPM, a galaxy schema was selected for the data warehouse architecture instead of a traditional single-star schema. In a standard star schema, different business processes with different structure are often forced into a single fact table, which can cause severe data duplication and slow down dashboard performance. The galaxy schema model solves this problem by maintaining multiple, independent fact tables that float side-by-side while seamlessly sharing conformed dimension tables. As displayed in Figure 4, this multidimensional model separates commercial sales logs from factory-floor production shifts, ensuring data integrity while allowing unified filtering across both domains in Google Data Studio.

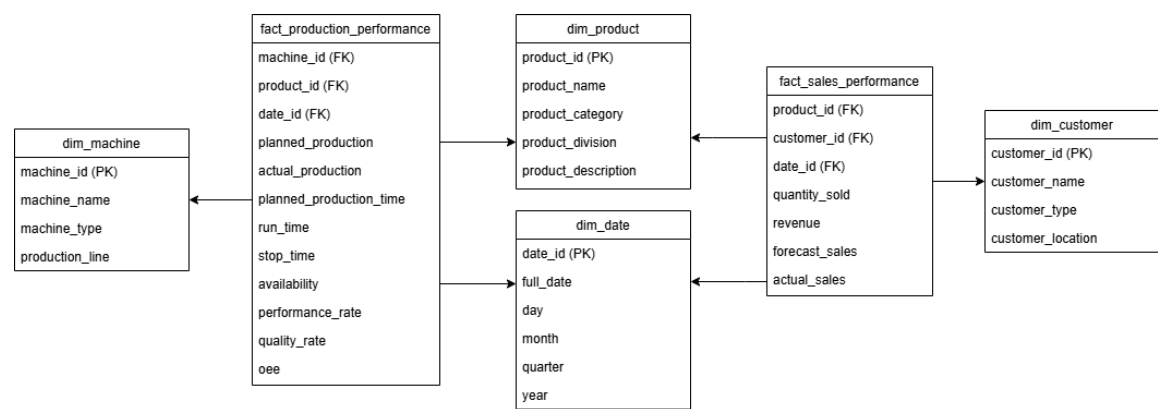


Figure 4 Galaxy Schema Model Design

7.1 Fact Table

Fact tables form the core of the multidimensional model, containing the quantitative, measurable metrics and foreign keys used to evaluate business performance. To isolate independent business activities and prevent reporting errors, the model utilizes two distinct fact tables, described in Table 1 below.

Table Name	Purpose	Connected Dimension Table
fact_sales_performance	Tracks financial turnover, customer demand, and sales target fulfillment.	dim_customer, dim_product, dim_date
fact_production_performance	Tracks factory-floor manufacturing volumes, equipment downtime, and asset efficiency.	dim_machine, dim_product, dim_date

Table 1 Fact Table Summary

The separation of these two fact tables ensures that independent operational metrics are evaluated accurately based on their specific business streams. The fact_sales_performance table records individual, numerical transactional data such as quantities sold and revenue to let top management check sales achievements against original customer forecasts.

Meanwhile, a key optimization in this schema is the consolidation of the transactional Production Data and Machine Performance entities into a single unified table: fact_production_performance. In the original operational requirements, a strict one-to-one relationship exists between these two entities, meaning each individual production record generates exactly one corresponding machine efficiency evaluation. Merging them removes the need for the Business Intelligence platform to continuously perform resource-intensive table joins in the background, maximizing dashboard query processing speeds in Google Data Studio.

7.2 Dimension Table

Dimension tables surround the central fact tables, providing the descriptive context, attributes, and filtering boundaries needed to analyze the quantitative metrics. The roles, types, and operational links of the four dimension tables utilized in this model are outlined in Table 2 below. .

Table Name	Type	Explanation
dim_customer	Standard	Descriptive table that stores comprehensive profile details regarding client accounts, such as company names, client classifications, and geographic locations.
dim_machine	Standard	Descriptive table that stores detailed technical characteristics of the factory equipment, including specific machine names, types, and their assigned physical production lines.
dim_product	Shared	A shared lookup table containing the standardized baseline data for all manufactured packaging items, tracking their names, categories, and operational divisions.
dim_date	Shared	Shared master calendar table that breaks down calendar days into standardized business timelines which enables the dashboard to filter and aggregate metrics by day, month, quarter, or year.

Table 2 Dimension Table Summary

Each dimension table is built into the galaxy schema to serve a different operational purpose. The dim_customer table is included to separate buyer account information so that the sales department can analyze performance patterns. It allows management to filter sales transactions cleanly, enabling them to discover which client categories or geographic locations generate the highest revenue and demand. Similarly, the dim_machine table exists to split factory equipment logs from unrelated commercial data. By tracking specific machine names, types, and lines, the factory operations manager can drill down into equipment issues or downtime patterns to investigate operational bottlenecks and floor productivity.

The remaining entities are designed as shared dimensions to eliminate data silos and unify the separate business streams across departments. The dim_product table acts as a master reference lookup that ensures that everyone across the manufacturing and sales teams utilizes identical, standardized baseline materials. Keeping this table shared allows management to track exactly how much of a specific product type is currently being produced versus how much is being sold. Finally, the dim_date table functions as a

universal architectural bridge that connects the entire data warehouse. It is shared so that a single date widget at the top of the final dashboard page in Google Data Studio can filter and sync both sales revenue and machine productivity by day, month, quarter, or year simultaneously.

7.3 User Impact

This multidimensional layout of the galaxy schema directly supports the daily operational workflows of different departmental teams by controlling how they interact with the data warehouse. For the front-end staff, the separation of tables allows them to upload, update, and manage their specific departmental logs within clean, isolated spreadsheet layers without risking accidental overwrites or data corruption from other sectors. For the data analyst, this shared structure simplifies the automated Extract, Transform, Load (ETL) pipeline, as data from separate departments can be cleaned and loaded into independent destination tables simultaneously without causing system downtime.

Most importantly, this architecture delivers a highly optimized data foundation to the final presentation layer for business decision-makers. By replacing cluttered, unified spreadsheets with shared dimension tables, top management, the finance department, and the factory operations manager can interact with separate, tailored performance dashboards that stay perfectly synchronized. When a user applies a single cross-functional filter such as a specific packaging product or calendar month, both dashboards update instantly. This streamlined layout removes technical complexity for the executive team, turning raw departmental inputs into immediate, data-driven decisions.

8.0 Conclusion

In conclusion, the development of the Business Intelligence (BI) dashboard system successfully addresses the operational monitoring challenges faced by PT. MPM. By following the structured waterfall methodology, the system requirements from various departments were effectively gathered, analyzed, and implemented. The raw and scattered transactional data sources originally mapped out in the Entity Relationship Diagram (ERD) were systematically refined through a clean Extract, Transform, Load (ETL) process managed by the data analyst. This automated data pipeline loaded the cleaned datasets into a centralized data warehouse using Google Sheets as the primary storage tool.

The resulting data warehouse layer utilizes an optimized, high-performance galaxy schema architecture. By keeping the commercial sales metrics and factory manufacturing logs in two separate fact tables, the system prevents data duplication and reporting errors, ensuring that calculations for revenue and Overall Equipment Effectiveness (OEE) remain completely accurate. At the same time, these independent tables are linked through shared conformed dimensions, the final presentation layer benefits from maximum query processing speeds without background lagging. This multidimensional model gives top management, the operations manager, and the finance department instant access to interactive performance visuals, enabling fast, data-driven strategic planning and enhancing the company's long-term operational efficiency.